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The Canadian Fourth Generation Atmospheric Global Climate Model (CanAM4). Part I: Representation of Physical Processes

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ABSTRACT *The Canadian Centre for Climate Modelling and Analysis (CCCma) has developed the fourth generation of the Canadian Atmospheric Global Climate Model (CanAM4). The new model includes substantially modified physical parameterizations compared to its predecessor. In particular, the treatment of clouds, cloud radiative effects, and precipitation has been modified. Aerosol direct and indirect effects are calculated based on a bulk aerosol scheme. Simulation results for present-day global climate are analyzed, with a focus on cloud radiative effects and precipitation. Good overall agreement is found between climatological mean short- and long-wave cloud radiative effects and observations from the Clouds and Earth's Radiant Energy System (CERES) experiment. An analysis of the responses of cloud radiative effects to variations in climate will be presented in a companion paper.*

RÉSUMÉ [Traduit par la rédaction] *Le Centre canadien de la modélisation et de l'analyse climatique (CCmaC) a mis au point la quatrième génération du modèle canadien de circulation générale de l'atmosphère (CanAM4). Le nouveau modèle comprend des paramétrisations physiques passablement modifiées comparativement à son prédécesseur. En particulier, le traitement des nuages, des effets radiatifs des nuages et des précipitations a été modifié. Les effets directs et indirects des aérosols sont calculés à l'aide d'un schéma d'aérosols en bloc. Nous analysons des résultats de simulation pour le climat général du jour présent en mettant l'accent sur les effets radiatifs des nuages et les précipitations. Nous trouvons un bon accord général entre la moyenne climatologique des effets radiatifs des nuages pour les courtes et les grandes longueurs d'onde et les observations de l'expérience CERES (Clouds and Earth's Radiant Energy System). Une analyse de la réponse des effets radiatifs des nuages aux variations du climat sera présentée dans un article connexe.*

KEYWORDS global climate model; cloud parameterization; convection; radiative transfer

1 Introduction

The fourth generation of the Canadian Atmospheric Global Climate Model (CanAM4), developed by the Canadian Centre for Climate Modelling and Analysis (CCCma), is based on the third generation Atmospheric General Circulation Model (AGCM3 McFarlane et al., 2006; Scinocca and McFarlane, 2004; Scinocca, McFarlane, Lazare, Li, and Plummer, 2008), which has been and continues to be widely used in climate research activities in Canada. However, CanAM4 differs substantially from AGCM3 in its treatment of a number of physical processes. In particular, CanAM4 includes prognostic representations of stratiform (layer) clouds and aerosols. In addition, treatments of radiative transfer, convection, and turbulent mixing

have been completely revised. These changes enable modelling experiments for established and newly emerging questions in climate research such as the role of aerosol/cloud interactions and feedbacks between atmospheric, biogeochemical, and oceanic processes in the climate system. CanAM4 is part of the Canadian Earth System Model (CanESM) and has recently been applied in numerous studies, including studies on carbon cycle feedbacks (Arora et al., 2011), seasonal forecasting (Merryfield et al., Unpublished manuscript; Fyfe et al., 2011) and attribution of climate change (Gillett, Arora, Flato, Scinocca, and von Salzen, 2012). Model results were provided to the Fifth Coupled Model Intercomparison Project (CMIP5; Taylor, Stouffer, and Meehl, 2011).

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Accurate simulations of global climate and climate change depend upon realistic model representations of clouds and their effects on radiative transfer and on amounts and distribution of heat, moisture, and chemical tracers in the atmosphere. For instance, all past climate assessment reports by the Intergovernmental Panel on Climate Change (IPCC) concluded that cloud feedbacks remain the largest source of uncertainty in climate sensitivity estimates (e.g., Forster et al., 2007). In CanAM4 and other models, simulated clouds are affected by radiation, transport, and a multitude of thermodynamic, microphysical, and chemical processes. Variations in simulated cloud processes and properties are substantial in space and time and are non-trivially related to radiative forcings from solar radiation, greenhouse gases, and short-lived climate forcers. In addition, interactions between atmospheric processes and land/ocean processes have considerable impacts on clouds. Effects of processes or model parameters on cloud properties are the focus of a considerable amount of scientific research.

The approach taken here is to analyze fundamental climatological model results, with an emphasis on clouds and precipitation. This provides a baseline for a subsequent analysis of responses of cloud radiative effects and precipitation to variations in climate in the second part of this paper.

This study is partly motivated by results from the Cloud Feedback Model Intercomparison Project (CFMIP). According to Williams and Webb (2009), a developmental version of this model (CCCma AGCM4) produced a particularly skillful representation of clouds for different cloud regimes, including those which are considered most important in a warming climate. Here, a more recent version of the model and improved satellite datasets are used.

The outline of the paper is as follows. A description of the modelling approach is given in Sections 2 and 3. Climatological mean results and variability for cloud radiative effects and precipitation are discussed in Section 5. Finally, Section 6 provides a brief summary.

2 Model description

a Overview of model changes

A summary of the main model features in CanAM4 and a comparison with the approach taken in the previous version of the model, AGCM3, are presented in Table 1. These features will be described in detail in the following sections.

b Advection of thermodynamic quantities and chemical tracers

As in AGCM3, horizontal advection is performed spectrally in CanAM4 while vertical advection employs rectangular finite elements defined for a hybrid vertical coordinate as described by Laprise and Girard (1990). Spectral transforms required for evaluation of quadratic products associated with advection are carried out on the usual quadratic Gaussian grid, which eliminates aliasing of unresolved spectral components arising from projecting quadratic terms into the spectral domain.

A second set of spectral transforms is used in order to facilitate evaluation of the physical processes on a reduced linear grid associated with the chosen spectral truncation. The linear grid has a resolution of 128×64 for T63. Performing the physics tendency calculations on the linear grid rather than the quadratic grid results in considerable computational savings because the linear grid contains less than half the number of grid points.

Table 1. Summary of main model features.

	AGCM3	CanAM4
Horizontal resolution	T47 or T63, double transform	As in AGCM3
Vertical grid	31 levels, top ≈ 1 hPa (≈ 50 km), $\Delta z \approx 100$ m near surface	As in AGCM3, 35 levels (enhanced near tropopause)
Topography	Optimal spectral (Holzer, 1996)	As in AGCM3
Advection	Hybridized moisture variable, spectral and semi-Lagrangian advection, global mass fixer	As in AGCM3, but hybridized tracer variables, local mass fixer, physics filtering (Lander and Hoskins, 1997)
Tracer horizontal diffusion	Leith	Laplace
Boundary layer turbulence	Based on bulk Richardson number (Abdella and McFarlane, 1996), dry non-local thermodynamic mixing	As in AGCM3, revised mixing lengths, moist non-local thermodynamic mixing
Land surface scheme	CLASS 2.7 (Verseghy 2000; Verseghy, McFarlane, and Lazare, 1993), predicted soil temperature and humidity	As in AGCM3
Gravity wave drag	Anisotropic orographic low-level drag (Scinocca and McFarlane, 2000)	As in AGCM3
Radiative transfer	4-band solar, 6-band terrestrial (Fouquart and Bonnel, 1980; Morcrette, 1984)	Correlated- k distribution and McICA
Clouds	Diagnostic cloud properties (McFarlane, Boer, Blanchet, and Lazare, 1992)	Prognostic cloud liquid water and ice, statistical cloud scheme, interactive with aerosols
Moist convection	Mass flux scheme for shallow and deep convection (Zhang and McFarlane, 1995)	As in AGCM3, mass flux scheme for deep convection, extended for aerosol chemistry. Separate shallow convection scheme (von Salzen, McFarlane, and Lazare, 2005)
Cloud base closure for deep convection	Diagnostic (Zhang and McFarlane, 1995)	Prognostic (Scinocca and McFarlane, 2004)
Atmospheric composition	Specified GHGs and aerosols	Specified GHGs, prognostic bulk aerosols

As in AGCM3, the vertical domain of CanAM4 extends from the surface to the stratopause region (1 hPa, approximately 50 km above the surface), but the vertical resolution is slightly higher in CanAM4 across the tropopause where more uniform resolution is employed. In the vertical, the domain is spanned by 35 layers. Layer depths increase monotonically with height from approximately 100 m at the surface to 3 km in the lower stratosphere.

CanAM4 employs two strategies that are used in combination to deal with the artifacts of overshoots and undershoots associated with spectral advection (Gibbs effect), which can induce negative concentrations of positive definite tracers. In the first, the hybrid variable approach adopted for the moisture variable in AGCM3 has been generalized and extended to tracers in CanAM4. The second strategy follows an approach first suggested by Lander and Hoskins (1997) in which both input fields and the output tendencies from the physics package are spatially filtered. Further details regarding the implementation of the second strategy may be found in Scinocca et al. (2008).

According to the hybrid variable approach, the quantity that is transported is the transformed variable

$$S = \begin{cases} \frac{q_0}{[1 + p \ln(q_0/q)]^{1/p}}, & \text{if } q < q_0, \\ q & \text{if } q \geq q_0, \end{cases} \quad (1)$$

with inverse transformation for $S < q_0$,

$$q = q_0 \exp\left[\frac{1 - (q_0/S)^p}{p}\right], \quad (2)$$

where q is the physical variable (i.e., specific humidity and chemical tracer mixing ratios), and q_0 and p are constants. This is a generalization of the hybrid variable transformation proposed by Boer (1995).

The use of this transformed variable alleviates, to a considerable extent, the undesirable overshoots and undershoots that can occur when spectrally transporting rapidly varying tracers (Merryfield, McFarlane, and Lazare, 2003). In particular, unphysical negative values of positive definite tracers are largely suppressed. One drawback of the hybrid transform approach involves mass conservation. Because the transformed variable S is now advected, it is spectrally conserved rather than the original variable q . However, by judiciously choosing q_0 and p , the degree of time- and globally averaged non-conservation can be effectively controlled. Therefore, q_0 and p are empirically assigned separately for each tracer according to a criterion that time-averaged global non-conservation errors do not exceed a certain threshold under present-day climate conditions. At each time step, global mass conservation of each hybrid tracer is also enforced in CanAM4 by applying local corrections for any residual mass changes which result from the transport of its hybrid form for static choices of q_0 and p in Eq. (1). As is evident from

theory and numerical experiment, the sign and magnitude of the correction necessarily depends on the given tracer distribution so that the correction will generally be different at each time step if the tracer concentrations are changing with time. The method is described in Appendix A.

c Aerosols

Different types of natural and anthropogenic aerosols are considered in CanAM4, including sulphate, black and organic carbon, sea salt, and mineral dust. Parameterizations for emissions, transport, gas-phase and aqueous-phase chemistry, and dry and wet deposition account for interactions with simulated meteorological variables in CanAM4 for each individual grid point and time step.

Similar to an earlier version of the model (Lohmann, von Salzen, McFarlane, Leighton, and Feichter, 1999), various processes are considered for sulphate. Monthly mean emissions of gas phase sulphur dioxide (SO_2) from anthropogenic and biomass burning sources are from Lamarque et al. (2010). In addition, climatologically representative emissions for non-explosive volcanoes (Dentener et al., 2006) are included. The model also accounts for production of SO_2 from oxidation of dimethylsulphide (DMS). For DMS, monthly mean emissions from the terrestrial biosphere (Spiro, Jacob, and Logan, 1992) and emissions based on climatological concentrations of DMS in sea water (Kettle et al., 1999) are used.

Under clear-sky conditions daytime production of sulphate (SO_4^{2-}) aerosol occurs via the oxidation of SO_2 by the hydroxyl radical (OH). At nighttime, SO_4^{2-} is produced by oxidation of SO_2 by the nitrate radical (NO_3). Chemical reactions in CanAM4 are summarized in Table 2.

The model accounts for in-cloud oxidation in deep and shallow convection, and stratiform clouds with hydrogen peroxide (H_2O_2) and ozone (O_3) as oxidants (von Salzen et al., 2000). The oxidation rates depend on the pH of the cloud water (Table 2), which is calculated from an ion balance for the dissolution of the chemical species SO_2 , ammonia (NH_3), nitric acid (HNO_3), and carbon dioxide (CO_2) in cloud droplets (von Salzen et al., 2000). For in-cloud oxidation in deep convection, the cumulus cloud fraction is calculated from the precipitation flux (Slingo, 1987).

Three-dimensional monthly averaged concentrations of OH, O_3 , H_2O_2 and NO_3 from the Model for Ozone and Related Chemical Tracers (MOZART; Brasseur et al., 1998) are used for aqueous and gas-phase chemical processes. Additionally, concentrations of NH_3 and ammonium (NH_4^+) from Dentener and Crutzen (1994) are used for the calculation of pH-dependent reaction rates in the clouds.

Primary particle emissions for black and organic carbon aerosol from anthropogenic and biomass burning sources are based on Lamarque et al. (2010). In addition, emissions of precursors for secondary organic aerosol are considered using monthly mean emissions (Dentener et al., 2006). Hydrophobic and hydrophilic black and organic carbon aerosol are considered, with a specified lifetime of $\tau = 24$ h for conversion

Table 2. Chemical reactions included in the model.

Reaction	Rate Coefficient or Equilibrium Constant ^b	Unit	Reference ^a
<i>Gas Phase Reactions</i>			
DMS + OH → SO ₂	$9.6 \times 10^{-12} \exp(-234/T)$	cm ³ molec ⁻¹ s ⁻¹	1
DMS + NO ₃ → SO ₂	$1.9 \times 10^{-13} \exp(500/T)$	cm ³ molec ⁻¹ s ⁻¹	2
SO ₂ + OH → H ₂ SO ₄	$a_0 \times m / (1 + a_0 \times m / a_\infty) \times 0.6^{[1. / (1 + \log(a_0 m) / a_\infty)^2]}$ with $a_0 = 3 \times 10^{-31} \times (300/T)^{3.3}$ and $a_\infty = 1.5 \times 10^{-12}$	cm ³ molec ⁻¹ s ⁻¹	3
<i>Equilibrium Reactions</i>			
SO ₂ (g) + H ₂ O(aq) ↔ SO ₂ (aq)	$1.23 \times 3120(1/T - 1/298)$	M atm ⁻¹	4
SO ₂ (aq) ↔ H ⁺ + HSO ₃ ⁻	$35.53(1/T - 1/298)$	M	4
HSO ₃ ⁻ ↔ H ⁺ + SO ₃ ²⁻	6.72×10^{-5}	M atm ⁻¹	4
O ₃ (g) + H ₂ O(aq) ↔ O ₃ (aq)	$29.44(1/T - 1/298)$	M atm ⁻¹	4
H ₂ O ₂ (g) + H ₂ O(aq) ↔ H ₂ O ₂ (aq)	6.4×10^8		4
<i>Aqueous Phase Reactions</i>			
S(IV) + H ₂ O ₂ → S(VI) + H ₂ O	$8 \times 10^4 \exp[-3650(1/T - 1/298)] (0.1 + [\text{H}^+])^{-1}$	M s ⁻¹	5
S(IV) + O ₃ → S(VI) + O ₂	$4.4 \times 10^1 \exp(-4131/T) + 2.6 \times 10^3 \exp(-966/T)[\text{H}^+]^{-1}$	M s ⁻¹	6

^aReferences: 1. Atkinson, Baulch, Cox, Hampson, and Troe (1989); 2. DeMoore et al. (1992); 3. Pham, Müller, Brasseur, Granier, and Mégie (1989); 4. Chameides (1984); 5. Martin (1984); 6. Maahs (1983)

^b T : temperature, m : air density in molec cm⁻³.

of hydrophobic to hydrophilic aerosol, representing aerosol aging processes (Croft, Lohmann, and von Salzen, 2005).

For dry sea salt aerosol, two separate log-normally distributed size modes with median radii $R_g = 0.209 \mu\text{m}$ and $R_g = 1.75 \mu\text{m}$ and geometric standard deviation $\sigma_g = 2.03$ are considered. Concentrations of hygroscopically grown sea salt particles in the first layer above the sea surface are calculated as a function of wind speed based on a relationship that was obtained from least-squares fitting of observations from the Joint Air-Sea Interaction (JASIN) experiment (Fairall, Davidson, and Schacher, 1983). Above this layer and over land, aerosol concentrations are determined by gravitational settling, dry and wet deposition, and transport.

The source flux for coarse ($R_g = 1.9 \mu\text{m}$, $\sigma = 2.15$) and accumulation mode ($R_g = 0.39 \mu\text{m}$, $\sigma = 2$) mineral dust is calculated based on the work of Martcorena and Bergametti (1995). It is proportional to the cube of the surface wind friction speed, including a gustiness adjustment, above a threshold value. It is additionally dependent on the fraction of exposed soil, the clay (accumulation mode) or silt (coarse mode) fraction in the originating soil and has a dependence on soil moisture similar to Fécan, Martcorena, and Bergametti (1999). A proportionality constant was determined by fitting the simulated deposition to the Dust Indicators and Records of Terrestrial and Marine Paleoenvironments (DIRTMAP) II database of climatological ice-core and ocean sediment dust deposition data (Kohfeld and Harrison, 1999). Similar to sea salt, once in the atmosphere, the dust is subject to transport, gravitational settling, and dry and wet deposition.

The parameterization of dry deposition in CanAM4 is based on specified deposition velocities for each type of aerosol and a further dependency on local conditions at the surface (Croft et al., 2005; Lohmann et al., 1999). Wet deposition fluxes from below- and in-cloud scavenging of aerosols depend on local rates of precipitation and conversion of cloud water to rainwater, respectively (Croft et al., 2005).

The approach of Dufresne, Quaas, Boucher, Denvil, and Fairhead (2005) is used in CanAM4 to calculate the cloud

droplet number concentration as a function of sulphate aerosol concentration. Following their approach, parameters in the parameterization of cloud droplet number were tuned in order to match simulated and observed results for the cloud droplet effective radius, giving

$$N_c = 60(\text{SO}_4^{2-})^{0.2}, \quad (3)$$

with N_c in units of droplets cm⁻³ and the concentration of sulphate, SO_4^{2-} , in units of $\mu\text{g m}^{-3}$. A lower bound of 1 droplet cm⁻³ is used for N_c . See Ma, von Salzen, and Cole (2010) for details.

The treatment of the radiative effects of aerosols in CanAM4 is described in Section 2f.

d Layer clouds

1 STATISTICAL CLOUD SCHEME

Similar to other models that employ statistical cloud schemes, statistical cloud properties in CanAM4 are diagnosed from grid-cell mean thermodynamic quantities. Following the basic framework that was first outlined by Mellor (1977), quasi-conserved thermodynamic variables are used, that is, liquid/ice water static energy ($h = c_p T_l + gz$) and total (non-precipitating) water specific humidity ($q_t = q_v + q_l + q_i$). Here, $T_l = T - (L_v q_l + L_s q_i) / c_p$ is the liquid water temperature, c_p the specific heat capacity at constant pressure, g the gravitational acceleration, z the height, T the temperature, L_v the latent heat of vapourization, and L_s the latent heat of sublimation. The specific humidities of liquid water, ice, and water vapour are q_l , q_i , and q_v , respectively. The variables h and q_t are directly derived from prognostic temperature, humidity, cloud condensate, and geopotential height in CanAM4.

Following Mellor (1977), the use of a truncated Taylor series for the saturation specific humidity leads to expressions for cloud parameters in terms of the variable

$$s = \frac{1}{2}(aq'_t - bT'_l),$$

with

$$a = \left(1 + \frac{(1 - f_i)L_v + f_i L_s}{c_p} \frac{\partial q_s}{\partial T} \Big|_{T=\bar{T}_l} \right)^{-1},$$

$$b = a \frac{\partial q_s}{\partial T} \Big|_{T=\bar{T}_l},$$

where the saturation specific humidity is q_s , and the ice fraction $f_i = q_i/(q_l + q_i)$. Bars indicate averaging over the grid cell volume and primed variables refer to deviations from the mean. In the model, f_i is calculated based on cloud liquid water and ice concentrations from the previous time step. If no cloud existed during the previous time step, an empirical, temperature-dependent fractional probability of liquid water (Rockel, Raschke, and Weyres, 1991) is used to initialize the cloud microphysical calculations.

Based on the assumption of a joint Gaussian probability distribution for h and q_t in each grid cell of the model, the mean α th moment of the cloud liquid/ice specific humidity and the variance of the cloud condensate distribution can be expressed in terms of s , that is,

$$\bar{q}_c^\alpha = \int_{-a\Delta\bar{q}/2}^{\infty} (a\Delta\bar{q} + 2s)^\alpha p(s) ds, \quad (4)$$

$$\bar{q}_c'^2 = \int_{-a\Delta\bar{q}/2}^{\infty} (a\Delta\bar{q} + 2s - \bar{q}_c)^2 p(s) ds, \quad (5)$$

with the cloud condensate specific humidity $q_c = q_l + q_i$, and the mean saturation deficit $\Delta\bar{q} = \bar{q}_l - q_s(\bar{T}_l)$.

The probability distribution in Eqs (4) and (5) is given by

$$p(s) = \frac{1}{\sqrt{2\pi}\sigma_s} \exp\left(-\frac{s^2}{2\sigma_s^2}\right),$$

with the variance

$$\sigma_s^2 = \frac{1}{4} (a^2 \bar{q}_l'^2 - 2ab \bar{q}_l' \bar{T}_l' + b^2 \bar{T}_l'^2). \quad (6)$$

Solutions to Eqs (4) and (5) can be obtained either analytically (e.g., Mellor, 1977) or numerically, depending on the value of α . The assumption of Gaussian probability distributions in this approach leads to qualitatively good agreement with observations (Larson et al., 2001). However, in order to account for deviations of probability distributions from Gaussian distributions, fitted results based on cloud-resolving model (CRM) output (Chaboureaud and Bechtold, 2002) are used to diagnose the condensate mixing ratios and cloud fraction,

$$C = \max\{0, \min[1, 0.5 + 0.36 \arctan(1.55Q_1)]\},$$

with

$$Q_1 = \frac{a\Delta\bar{q}}{\sigma_s}.$$

Unfortunately, results for other moments of the cloud condensate distribution are not presently available from CRM simulations.

The treatment of variance for s (Eq. (6)) is based on the approach by Chaboureaud & Bechtold (2005). According to this approach, the total variance is decomposed into local turbulent (subscript t) and convective (subscript c) contributions, that is,

$$\sigma_s^2 = \sigma_t^2 + \sigma_c^2,$$

with

$$\sigma_t = c_\sigma l_c \left| a \frac{\partial \bar{q}_l}{\partial z} - \frac{b}{c_p} \frac{\partial \bar{h}}{\partial z} \right|,$$

with $c_\sigma = 0.2$. A mixing length $l_c = 600$ m in the cloudy part of the free troposphere is assumed to account for sub-grid-scale effects of radiative cooling at cloud top and buoyancy sorting on mixing, following Chaboureaud and Bechtold (2005).

The diagnostic treatment of the convective contribution σ_c that was originally proposed by Chaboureaud and Bechtold (2005) has been slightly modified to account for the aging of convectively generated perturbations caused by gravity waves. For typical horizontal grid sizes in global climate models (i.e., a few hundred kilometres), sources of variance from convection cannot be resolved. Gravity wave activity associated with convective events is expected to cause dissipation of the convectively injected variance. In order to account for this effect, the following approach is used:

$$\sigma_c(z, t) = \begin{cases} 3 \times 10^{-3} |M_c| a^{-1}, & \text{if } M_c \neq 0, \\ \sigma_c(z, t - \Delta t) \exp(-\Delta t/\tau_c), & \text{if } M_c = 0, \end{cases} \quad (7)$$

with convective mass flux M_c . As a modification to the original parameterization by Chaboureaud and Bechtold (2005), the variance decays over time on a time scale of $\tau_c = 6$ h in Eq. (7), which is within the range of results from studies on the decay of deep convection (e.g., Khairoutdinov and Randall, 2002).

The statistical cloud scheme is also used to calculate humidity profiles in parameterizations of clear-sky radiative transfer. The mean specific humidity in the clear portion of the grid cells after adjustment to thermodynamic equilibrium is given by

$$q_{v,\text{clear}} = \frac{\bar{q}_v - C q_s}{1 - C}.$$

2 CLOUD MICROPHYSICS

A prognostic microphysics scheme is used for simulations of stratiform clouds. The governing equations for the mass mixing ratios of water vapour, cloud liquid water, and cloud ice are based on the approach by Lohmann and Roeckner (1996) and Lohmann (1996). Basic microphysical processes in the scheme are similar to those found in other state-of-the-art global models.

Figure 1 and Table 3 provide an overview of microphysical processes in the cloud scheme.

Condensation and evaporation (Q_{cnd}) are treated as an instantaneous adjustment of the thermodynamic properties in

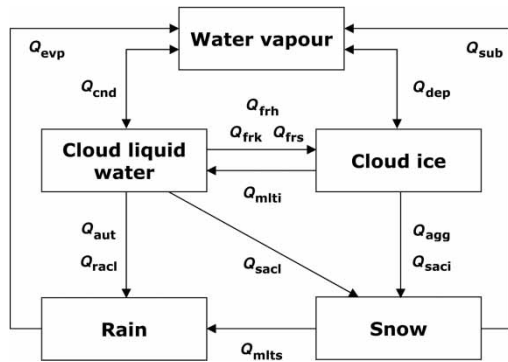


Fig. 1 Microphysical processes in layer clouds in CanAM4. See Table 3 for an explanation of terms.

the grid cells to equilibrium as given by the statistical cloud scheme.

The relative humidity in ice clouds may be either lower or higher than saturation over ice. Time scales for water vapour deposition (Q_{dep}) and sublimation (Q_{sub}) in ice clouds typically vary from minutes to hours. Consequently, relative humidities are affected by cloud dynamics and other cloud processes with finite time scales (e.g., Korolev and Isaac, 2006). For simplicity and numerical efficiency, non-equilibrium effects are omitted in CanAM4 for the deposition of water vapour onto ice crystals and sublimation of ice crystals.

Parameterizations of autoconversion of cloud droplets to rain (Q_{aut}) and accretion of cloud droplets by rain (Q_{racl}) are based on results of cloud-resolving model simulations (Khairoutdinov and Kogan, 2000). The mean autoconversion rates in the model grid cells are calculated using probability distributions of cloud water from the statistical cloud scheme by using a look-up table containing a numerical solution of Eq. (4), with $\alpha = 2.47$.

The original parameterization of autoconversion by Khairoutdinov and Kogan (2000) was modified for application in CanAM4. Firstly, the conversion rate is scaled up by a factor of 1.3, which provides better agreement of global mean model results and observations for cloud liquid water path and cloud radiative effects. The sensitivity of model

results to this factor has been analyzed by Cole, Barker, Loeb, and von Salzen (2011). Secondly, the second indirect aerosol effect is not considered in the model given substantial uncertainties that are associated with the representation of this effect in models (e.g., Lohmann and Feichter, 2005). Hence, a global constant cloud droplet number concentration ($N_c = 50 \text{ cm}^{-3}$) is used in the parameterization for autoconversion of cloud water into rain instead of Eq. (3). Although these changes may appear to be substantial, the approach is justifiable in the view of large overall uncertainties in parameterizations of autoconversion (Wood, 2005) and the uncertain role of cloud droplet concentrations in interactions between cloud microphysical and dynamical processes (Ackerman, Kirkpatrick, Stevens, and Toon, 2004).

The net conversion rate for autoconversion and accretion ($Q_{\text{aut}} + Q_{\text{racl}}$) is simulated by using an iterative semi-implicit method.

Yuan, Fu, and McFarlane (2006) identified inconsistencies in the treatments of accretion processes in the microphysics scheme by Lohmann and Roeckner (1996) which was used in an early developmental version of CanAM4. This was subsequently addressed in the development of CanAM4 by using the approach of Rotsteyn (1997) for accretion of cloud droplets by snow (Q_{saci}) and accretion of ice crystals by snow (Q_{saci}). For the latter, the collection efficiency that was proposed by Levkov, Rockel, Kapitza, and Raschke (1992) is used.

As a further modification to the original cloud scheme by Lohmann and Roeckner (1996), evaporation of rain (Q_{evp}) is also treated according to the approach of Rotsteyn (1997) in CanAM4.

Rain and snow contents are diagnostically calculated from the corresponding precipitation fluxes $R_{r,s}$ for calculations of Q_{racl} and Q_{sub} , that is,

$$q_{r,s} = \frac{R_{r,s}}{\rho v_{r,s}},$$

with the density of air, ρ , and terminal fall velocities $v_{r,s}$ for rain (subscript r) and snow (subscript s). The terminal fall velocities that are used in these calculations represent the mean fall velocities for spectra of droplet and crystal sizes (Rotsteyn, 1997).

Sedimentation of ice crystals in CanAM4 is parameterized in terms of the mean ice crystal diameter, which is determined from the cloud ice content (Lohmann and Roeckner, 1996; Murakami, 1990). However, considerable uncertainty is associated with this process owing to the wide range of sizes and shapes of ice crystals in the atmosphere (Heymsfield, 2003). Additional uncertainties in ice crystal contents arise from complex interactions of sedimentation with other microphysical processes and mixing (Kay, Baker, and Hegg, 2006). In order to account for these uncertainties and to reduce systematic model biases for high clouds, the ice terminal fall velocity in CanAM4 is multiplied by a factor of $a_I = 6000$ instead of $a_I = 700$ in the original approach by Murakami (1990).

Table 3. Summary of cloud microphysical processes.

Variable	Description
Q_{agg}	Aggregation of ice crystals
Q_{aut}	Autoconversion of cloud droplets to rain
Q_{cnd}	Condensation and evaporation of cloud liquid water
Q_{dep}	Deposition of water vapour and sublimation of cloud ice
Q_{evp}	Evaporation of rain
Q_{frh}	Homogeneous freezing of cloud droplets
Q_{frs}	Stochastic and heterogeneous freezing of cloud droplets
Q_{frk}	Contact freezing of cloud droplets
Q_{mlti}	Melting of cloud ice
Q_{mlts}	Melting of snow
Q_{racl}	Accretion of cloud droplets by rain
Q_{saci}	Accretion of ice crystals by snow
Q_{saci}	Accretion of cloud droplets by snow
Q_{sub}	Sublimation of snow

Another modification in CanAM4 is the scaling of the aggregation rate for ice crystals to snow (Q_{agg}). Lohmann (1996) used a formulation of the aggregation time scale that was originally proposed by Murakami (1990) but multiplied the time scale by a factor of $\gamma = 220$. In CanAM4, the same formulation is retained with $\gamma = 50$, which produces good agreement between simulation results and satellite observations for clouds.

The representation of the unresolved precipitation flux in microphysical calculations is described in Appendix B. The approach accounts for partial overlap of different cloud layers in the vertical.

e Moist convection

CanAM4 includes separate parameterizations for deep and shallow convection. Both parameterizations of convection use the same input profiles of temperature, moisture, and chemical tracer mixing ratios, which were output from the prognostic cloud scheme rendering them statically stable and at most fully saturated. Both schemes are permitted to be active in the same grid cells at any time within specific physical constraints for each scheme (von Salzen et al., 2005; Xie et al., 2002).

Similar to AGCM3, the cumulus parameterization of Zhang and McFarlane (1995) is used to represent the effects of deep convection (hereafter denoted by ZM) in the model. The ZM-parameterization is a bulk mass flux scheme which includes a representation of convective scale motions. It is designed to account for the effects of convective updrafts and downdrafts from evaporation of rain. As a modification of the original approach, the ZM-parameterization in CanAM4 is applied only to cumulus cloud ensembles with maximum cloud top heights above the ambient freezing level as predicted by the parameterization. This effectively limits the application of the parameterization to cumulonimbus and cumulus congestus types of clouds. A prognostic closure, based on convectively available potential energy (CAPE), is used (Scinocca and McFarlane, 2004).

Vertical momentum transfer in deep convection is included as a vertical flux term in the form:

$$\left. \frac{\partial(\rho \mathbf{V})}{\partial t} \right|_{\text{conv}} = - \frac{\partial[M_c(\mathbf{V}_c - \mathbf{V})]}{\partial z},$$

where $\mathbf{V}$ is the large-scale horizontal velocity, and $\mathbf{V}_c$ is the corresponding convective scale horizontal velocity. This quantity is determined by solving the convective scale momentum budget equation in the form

$$\frac{\partial(M_c \mathbf{V}_c)}{\partial z} + D \mathbf{V}_c - E \mathbf{V} = -f_c(\nabla_H p)_c, \quad (8)$$

where f_c is the fractional area covered by convective-scale updrafts, M_c is the deep convective updraft mass flux, D and E are, respectively, mass entrainment and detrainment rates

such that

$$\frac{\partial M_c}{\partial z} + D - E = 0.$$

These quantities are determined in the ZM-parameterization.

The pressure gradient force is parameterized following Gregory, Kershaw, and Inness (1997) as

$$-f_c(\nabla_H p')_c = \eta M_c \frac{\partial \mathbf{V}}{\partial z}. \quad (9)$$

With this parameterization it is easily shown that the convective scale horizontal velocity is a weighted average of the large-scale flow and its value in the absence of a convective scale pressure-gradient force,

$$\mathbf{V}_c = (1 - \eta)\mathbf{V}_0 + \eta \mathbf{V},$$

where $\mathbf{V}_0$ is the solution to Eq. (8) with the right-hand side set to zero.

This implies that the tendency associated with the cumulus momentum transfer is given by the value determined by ignoring the pressure-gradient force scaled by the factor $1 - \eta$. This relationship is used in implementing the parametrization. Gregory et al. (1997) suggested $\eta = 0.7$ based on cloud resolving model results and that is the default value used in CanAM4. However, Zhang and Wu (2003) have suggested a smaller value ($\eta = 0.55$) and more recently Romps (Unpublished manuscript) has argued that the formulation in Eq. (9) may produce non-physical results in some cases and that the choice $\eta = 0$ as in Schneider and Lindzen (1976) is preferable. Preliminary results show that CanAM4 simulations with $\eta = 0$ are no worse than with the default value. However, Romps (Unpublished manuscript) also advocated using a drag-law formulation for the convective scale momentum flux to account for situations with strong shear in the large-scale flow. Such a formulation has not yet been tested in CanAM4.

Effects of shallow convection are parameterized following von Salzen and McFarlane (2002) and von Salzen et al. (2005). In the parameterization, parcels of air are lifted from the planetary boundary layer (PBL) into the layer above the PBL. Shallow cumulus clouds are formed once the parcels reach the level of free convection (LFC), at which the parcels become positively buoyant. Above the LFC the parcels are modified by entrainment of environmental air into the ascending top of the cloud and also by organized entrainment at the lateral boundaries of the cloud. The cloud-top mixing produces horizontal inhomogeneities in cloud properties and vertical fluxes which are parameterized using joint probability density distributions of total water and moist static energy. The initial growth phase of the cumulus cloud is assumed to be terminated when its top reaches its maximum level. The growth phase is followed by

instantaneous decay, with complete detrainment of cloudy air into the environment.

Tendencies of thermodynamical and chemical tracers in the shallow cumulus scheme are calculated based on continuity equations for mass, dry static energy, moisture, chemical tracer mixing ratios, and vertical momentum. The mixing parameters used in CanAM4 are consistent with the parameters used in the experiment DECORE_B, as described by von Salzen and McFarlane (2002). A parameterization of autoconversion has been included in the scheme in order to account for the effects of drizzle formation in shallow cumulus clouds following Lohmann and Roeckner (1996).

For implementation of the shallow convection scheme in CanAM4, the cloud base closure condition proposed by Grant (2001) is used. This approach is based on a simplified turbulent kinetic energy (TKE) budget for the convective boundary layer. It is based on the assumption of steady mean boundary layer flow by omitting effects of pressure perturbations and shear on the TKE. Under the assumption that the buoyancy flux can be approximated by a linear function of height and by further assuming that the vertical flux of TKE at cloud base is proportional to the cloud base mass flux, a simple expression for the cloud base mass is obtained,

$$M_b = \rho \left[\frac{1}{2} (1 - \alpha_g) - A_\varepsilon \right] w_*, \quad (10)$$

where, for $\overline{w'\theta_v^s} > 0$,

$$w_* = \left(\frac{g \overline{w'\theta_v^s} z_i}{\theta_v} \right)^{\frac{1}{3}}, \quad (11)$$

is the sub-cloud layer convective velocity scale, θ_v is the mean virtual potential temperature of the mixed layer, $\overline{w'\theta_v^s}$ is the turbulent flux of virtual potential temperature at the surface, g is the gravitational acceleration, and ρ is the density of air. The variable z_i is the depth of the mixed layer, which is defined as that level in the model at which the gradient Richardson number exceeds $Ri = 1$. $\alpha_g = 0.2$ and $A_\varepsilon = 0.37$ in Eq. (10) are constants (Grant, 2001).

The parameterization of shallow convection is invoked only if the cloud tops, as predicted by this parameterization, are below the ambient freezing level.

f Radiation

1 OPTICAL PROPERTIES FOR GASES, CLOUDS, AEROSOLS, AND THE SURFACE

The absorption by gases in the atmosphere is parameterized using the Correlated- k Distribution (CKD) method (Li and Barker 2005) which replaces the band-mean absorption used in AGCM3 (Fouquart and Bonnel, 1980; Morcrette, 1984). The method upon which CKD is built, effectively sorts the gaseous absorption coefficients (k), which can vary greatly

with wavenumber, over a particular wavenumber range into much smoother cumulative probability distributions. These sorted probability distributions form a cumulative probability space (CPS). Doing so greatly reduces the number of radiative transfer calculations while maintaining good accuracy relative to computationally expensive benchmark radiative transfer calculations over the rapidly varying k 's. The CKD approach also improves the ability to model the overlapping absorption by more than one gas, especially relative to the method used in AGCM3. The gases accounted for in the CKD, which has four wavenumber intervals for the shortwave and nine intervals for the longwave, are listed in Table 4. Methods are described in Li and Barker (2005) to compute the combined absorption efficiently and accurately when there is more than one gas present. Further efficiencies are found by noting that for some integration points along the k distributions the gaseous absorption is very large relative to other absorbing and scattering constituents in the atmosphere (e.g., clouds and aerosols) and so the radiative transfer calculations can be greatly simplified by neglecting scattering and only considering absorption. In Table 4, two types of CPS intervals are listed: the *major* intervals are those for which the absorption by gases is small enough that scattering should be included in the radiative transfer calculations whereas for the *minor* intervals only absorption is included in the radiative transfer calculations.

In contrast to the optical properties of the gases, those for cloud particles and aerosols vary relatively slowly with wavenumber. Therefore, the parameterizations of these optical properties in CanAM4 are appropriately weighted mean values over each of the wavenumber bands listed in Table 4.

The cloud optical properties, specific extinction, single scattering albedo, and asymmetry parameter, are parameterized for each CKD band as a function of particle size and concentration. For liquid cloud particles the effective radius is computed assuming that the drop size is a gamma distribution (Ramaswamy and Li, 1996),

$$r_{e,l} = f \left(\frac{3\rho_l q_l}{4\pi\rho_l C(N_c \times 10^6)} \right)^{1/3}, \quad (12)$$

where q_l is the specific humidity of cloud liquid water, ρ_l is the density of water, C is the cloud fraction (Eq. (7)), and $f = 1.4$. For ice cloud particles their effective radius is computed using Lohmann and Roeckner (1996):

$$r_{e,i} = 83.8 \times 10^{-6} \left(\frac{10^3 \rho_i q_i}{C} \right)^{0.216}, \quad (13)$$

where q_i is the concentration of cloud ice. This effective radius is then related to a *generalized effective size* to account for non-spherical ice particles used to compute ice optical properties (Fu, 1996),

Table 4. The band spectrum ranges, absorbers, and the number of intervals for CPS. The intervals for CPS are divided into two categories, major and minor. See text for details.

				Number of interval in CPS	
Band	(cm ⁻¹)	Sub-band (cm ⁻¹)	Absorber	major	minor
<i>Solar</i>					
1	14500–50000	50000–42000: UVC	O ₂ , H ₂ O	6	3
		42000–37400: UVC			
		37400–35700: UVC			
		35700–34000: UVB			
		34000–32185: UVB, UVA			
		32185–30300: UVA			
		30300–25000: UVA			
		25000–20000: PAR			
2	8400–14500	20000–14500: PAR	H ₂ O, O ₃ , O ₂	4	4
3	4200–8400		H ₂ O, CO ₂ , CH ₄	6	4
4	2500–4200		H ₂ O, CO ₂ , CH ₄	4	9
Total				20	20
<i>Infrared</i>					
1	2200–2500		H ₂ O, CO ₂ , N ₂ O	1	5
2	1900–2200		H ₂ O, N ₂ O	1	1
3	1400–1900		H ₂ O	2	3
4	1100–1400		H ₂ O, N ₂ O, CH ₄ , CFC12	5	5
5	980–1100		H ₂ O, O ₃ , CO ₂ , CFC11, CFC12	2	4
6	800–980		H ₂ O, CO ₂ , CFC11, CFC12	3	0
7	540–800		H ₂ O, CO ₂ , N ₂ O, O ₃	3	7
8	340–540		H ₂ O	6	3
9	0–340		H ₂ O	4	6
Total				27	34

$$D_{ge} = \frac{8}{3\sqrt{3}} r_{e,i}. \quad (14)$$

With the cloud water contents and effective particles sizes, the cloud optical properties for each band in the CKD are computed for liquid cloud particles at solar (Dobbie, Li, and Chýlek, 1999) and infrared (Lindner and Li, 2000) wavenumbers and for ice cloud particles at solar (Fu, 1996) and infrared (Fu, Yang, and Sun, 1998) wavenumbers.

Several species of aerosols are radiatively active in CanAM4. Under the assumption that ammonium sulphate ((NH₄)₂SO₄) dominates the chemical composition of sulphate aerosols, their optical properties are parameterized using Li, Wong, Dobbie, and Chýlek (2001). Sea salt and mineral dust optical properties are parameterized using Lesins, Chýlek, and Lohmann (2002) while the optical properties of black and organic carbon are parameterized using Bäumer, Lohmann, Lesins, Li, and Croft (2007). Volcanic aerosols in the stratosphere are assumed to be distributed between a climatological mean tropopause and 10 hPa and to be composed of 25% water (H₂O) and 75% sulphuric acid (H₂SO₄). A log-normal distribution with a geometric radius of 0.1336 μm (effective radius of 0.35 μm) and effective variance of 0.46978 μm² is assumed (Hansen and Travis, 1974). Their optical properties are computed using Mie computations for each CKD band.

The parameterizations of surface albedo and emissivity over land and sea ice are largely unmodified from AGCM3 (see Section 2i). For ocean solar surface albedo, CanAM4 uses

the scheme of Jin, Charlock, and Rutledge (2002), which is dependent on solar zenith angle and surface wind speed, and accounts for the direct and diffuse components of the incident solar radiation. This scheme was modified to account for the effect of white-caps forming at relatively high wind speeds (>15 m s⁻¹; Monahan and MacNiocaill, 1986), which can enhance the surface albedo (Li et al., 2006). The ocean surface emissivity uses the broadband formulation of Hansen et al. (1983).

2 RADIATIVE TRANSFER

There are significant differences between the radiative transfer solver in AGCM3 and that used in CanAM4. The latter has the ability to use either the deterministic solvers described in Li (2002); Li, Dobbie, Räisänen, and Min (2005); Li and Barker (2005) or the Monte Carlo Independent Column Approximation (McICA; Barker et al., 2008; Pincus, Barker, and Morcrette, 2003). Although both schemes share the ability to account for radiative transfer through overlapping and horizontally inhomogeneous clouds, only the implementation of McICA will be described here.

The McICA solves the radiative transfer for each grid cell in CanAM4 by randomly sampling unresolved structure while systematically sampling each integration point in the CKD, thereby generating an unbiased estimate of grid cell–mean radiative fluxes. This requires, in addition to the optical properties described in the previous section, a radiative transfer solver and a method to model the unresolved structure, namely clouds. The radiative transfer solvers, one for solar

and one for the infrared, are effectively those described in Li et al. (2005) and Li (2002) although they have been simplified to solve the radiative transfer only through overcast homogeneous clouds in multiple layers because this is all that is required for McICA. The unresolved cloud structure is provided by a stochastic cloud generator (Räisänen, Barker, Khairoutdinov, Li, and Randall, 2004).

For solar radiation, a two-stream solver, along with a delta-Eddington approximation, is used to calculate radiative transfer in the atmosphere, which leads to a numerically efficient approach that has a linear dependence on the number of vertical levels. The effect of atmospheric spherical curvature and refraction on the effective pathlength is accounted for by adjusting the solar zenith angle using

$$\frac{1}{\mu_e} = \frac{24.35}{2\mu_0 + \sqrt{498.5225\mu_0^2 + 1}},$$

where μ_0 is the cosine of solar zenith angle and μ_e is the cosine of effective solar zenith angle (Li and Shibata, 2006).

For infrared radiation, a two-stream solver is also used along with a methodology to account for the scattering by cloud and aerosol particles efficiently (Li, 2002). In the CKD model, wavenumber 2500 cm^{-1} delineates the solar and infrared regions for the application of each of the radiative transfer solvers. However, there is approximately 12 Wm^{-2} of solar radiation incident at the top of the atmosphere for wavenumbers less than 2500 cm^{-1} . This solar radiation is used as an upper boundary condition for the downward fluxes in the infrared radiative transfer calculations (Li, Curry, Sun, and Zhang, 2010).

Two aspects of the unresolved cloud structure are accounted for in the CanAM4 radiative transfer calculations: the horizontal variability of cloud condensate and the vertical overlap of cloud. The cloud condensate is assumed to follow a gamma distribution, based on previous studies (e.g., Barker, 1996):

$$p = \frac{1}{\Gamma(\nu)} \left(\frac{\nu}{\bar{q}_c} \right)^\nu \bar{q}_c^{\nu-1} e^{-\nu \bar{q}_c / \bar{q}_c^2}, \quad (15)$$

where $\Gamma(\nu)$ is the gamma function, and $\nu = \bar{q}_c^2 / \overline{q_c'^2}$, where $\bar{q}_c$ and $\overline{q_c'^2}$ are the mean and the variance of the cloud water content, respectively. The parameter ν in Eq. (15) is calculated using $\bar{q}_c$ and $\overline{q_c'^2}$ from the Gaussian probability distributions of quasi-conserved thermodynamic variables in the statistical cloud scheme (Section 2d1). No variability is assumed in cloud particle size (i.e., it is horizontally uniform in a model layer) although this assumption can be relaxed.

Cloud vertical overlap is modelled using Hogan and Illingworth (2000) and Räisänen et al. (2004),

$$C_{k,l} = \alpha_{k,l} C_{k,l}^{\max} + (1 - \alpha_{k,l}) C_{k,l}^{\text{ran}}, \quad (16)$$

where $C_{k,l}$ is the vertically projected cloud fraction for two layers k and l , $C_{k,l}^{\max} = \max(C_k, C_l)$ (maximum overlap), and

$C_{k,l}^{\text{ran}} = C_k + C_l - C_k C_l$ (random overlap). The variable $\alpha_{k,l}$ is the overlap parameter which is given as:

$$\alpha_{k,l} = \exp\left(-\int_{z_k}^{z_l} \frac{dz}{L(z)}\right), \quad (17)$$

where $L(z)$ is the decorrelation length and z is the height. Because the layer cloud fractions C are computed in CanAM4, it is necessary to set $L(z)$ to specify the vertical overlap of clouds. Based on analysis of observations and output from cloud resolving models (Barker et al., 2008; Hogan and Illingworth, 2003; Pincus, Hannay, Klein, Xu, and Hemler, 2005), $L(z)$ was set to 2 km for cloud fraction and 1 km for cloud water content.

g Turbulence

Turbulent transfer of scalar quantities in the boundary layer involves both local, down-gradient transfer processes and non-local processes that are often associated with the occurrence of upward heat and moisture fluxes at the surface.

Non-local mixing of heat, moisture, and chemical tracers occurs in simulations with CanAM4 if the direction of the buoyancy flux at the surface is upward. The approach is based on the assumption that non-local effects can be represented by relaxing local values of liquid/ice water static energy, total water, and other quasi-conserved scalar quantities in the boundary layer, $\bar{\chi}$, to a vertically homogeneous reference state, $\bar{\chi}_r$, over a specified time period, τ , that is,

$$\left. \frac{\partial \bar{\chi}}{\partial t} \right|_{\text{nl}} = \frac{1}{\tau} (\chi_r - \bar{\chi}).$$

Similar to AGCM3, the reference state is determined by assuming that the vertical flux must vanish at the top of the mixing region, that is,

$$\chi_r = \frac{1}{1 - \sigma_t} \int_{\sigma_t}^1 \bar{\chi} d\sigma + \frac{g\tau p_s \overline{w'\chi'}}{p_s(1 - \sigma_t)},$$

where $\sigma_t = p_t/p_s$ is the sigma coordinate for the top of the mixing region. The variables p_s and ρ_s are, respectively, the pressure and density of the air at the surface, and $\overline{w'\chi'}$ is the surface flux for χ .

The top of the mixing region is determined iteratively by strapping layers together, beginning with the bottom layer alone and adding additional layers as long as the computed value of the virtual potential temperature in the topmost layer of the mixing region is larger than the ambient value in the layer directly above the top one in the mixing region. It can be shown that the treatment of non-local mixing in CanAM4 corresponds to the standard encroachment formula for growth of a convectively driven mixed layer if the temporal variation of the surface flux is negligible (McFarlane et al., 2006).

The mixing adjustment time scale τ is based on the eddy turnover time scale (Abdella and McFarlane, 1997), that is,

$$\tau = \frac{\alpha_t z_t}{\max[(0.6w_*^2 + u_*^2)^{1/2}, 0.03]},$$

with the convective velocity scale w_* , friction velocity u_* , $\alpha_t = 1$, and

$$z_t = \frac{p_s}{g} \int_{\sigma_t}^1 \rho^{-1} d\sigma.$$

Once vertical profiles for liquid/ice water static energy and total water are determined, the statistical cloud scheme is used to determine profiles of temperature, specific humidity, and cloud condensate.

The local down-gradient transfer of momentum, liquid/ice water static energy, total water, and prognostic trace constituents associated with turbulent transfer are accounted for by using diffusivities which are functions of the vertical wind shear and the local gradient Richardson number. The formulation used is qualitatively similar to that of AGCM2 (McFarlane et al., 1992, 2006). Eddy diffusivities for momentum, heat, and tracers are of the form

$$K = l^2 \left| \frac{\partial \mathbf{V}}{\partial z} \right| f(\text{Ri}),$$

where $\mathbf{V}$ is the horizontal wind vector and z the distance above the terrain. The factor $f(\text{Ri})$ depends on the gradient Richardson number Ri (McFarlane et al., 1992).

A unified formulation for the mixing length is used in CanAM4 for consistent treatment of local mixing under clear- and cloudy-sky conditions. It combines features of the approach that is used in AGCM3 with the statistical cloud scheme in CanAM4.

In the absence of clouds, the mixing length in the convectively mixed layer is given by

$$l = \max\left(\frac{l_u l_d}{l_u + l_d}, l_{\min}\right),$$

where l_u and l_d are length scales for upward and downward turbulent transfer, respectively (Lenderink and Holtslag, 2004). The minimum mixing length is given by

$$l_{\min} = \max\left(\frac{l_m l_w}{l_m + l_w}, l_0\right),$$

where $l_m = 75$ m, $l_w = 0.5l_u$, and $l_0 = 10$ m. In the atmosphere, l_u and l_d will generally depend on the stability of the atmosphere and the efficiency of non-local mixing. However, the

following simple approximations are used in CanAM4:

$$l_u = \kappa z,$$

$$l_d = \kappa(z_t - z),$$

with the von Kármán constant κ and the height above the surface, z .

Above the convectively mixed layer, it is assumed that the mixing length decreases monotonically with height according to

$$l = l_0 + (l_{\min} - l_0) \left(\frac{\sigma}{\sigma_t} \right)^2,$$

if no clouds are present.

Under cloudy conditions, a mixing length of $l_c = 600$ m in the free troposphere is assumed, as described in Section 2d1. For all-sky conditions, the mixing length is calculated from a linear combination of l and l_c ,

$$l_a = (1 - C)l + Cl_c,$$

with cloud fraction C .

The treatment of mixing in CanAM4 yields realistic profiles of water and heat in CanAM4 (e.g., Zhu et al., 2005).

h Surface fluxes

Surface exchanges of heat, moisture, and momentum follow the treatment of Abdella and McFarlane (1996). The approach is based on the similarity theory of Monin and Obukhov (1954) according to which the surface wind stress, sensible heat flux, and moisture flux, are related to the wind, temperature, and specific humidity near the top of the surface layer through friction velocity, temperature, and humidity scales.

CanAM4 accounts for the effects of wind gustiness on surface fluxes of momentum, heat, moisture, and tracers based on an approach that was proposed by Redelsperger, Guichard, and Mondon (2000). For this, an effective absolute wind speed V_{eff} is used in the calculations of surface fluxes, with

$$V_{\text{eff}} = \sqrt{U^2 + V^2 + V_1^2 + V_2^2},$$

where (U, V) is the wind speed in the first model layer and

$$V_1 = \beta w_*$$

$$V_2 = \log(1 + 6.69P_c - 0.476P_c^2),$$

with w_* from Eq. (11), $\beta = 0.55$, and the convective surface precipitation rate P_c (in cm d^{-1}). V_1 and V_2 refer to contributions of gustiness from boundary layer free convection and deep convection, respectively.

i Land and ocean surface

Similar to AGCM3, the calculation of energy and moisture fluxes at the land surface is carried out within the Canadian Land Surface Scheme (CLASS) module that was first introduced in AGCM3 (McFarlane et al., 2006). Development of the CLASS model started in the late 1980s. The module has undergone certain modifications since then. The basic physics underlying the scheme is outlined in two papers, Verseghy (1991) and Verseghy et al. (1993), and the changes implemented over the following decade are described in Verseghy (2000). The version of CLASS currently used in CanAM4 is referred to as version 2.7. A brief outline of its structure is provided below.

Each land surface grid cell treated by CLASS can have up to four subareas: bare soil, vegetation-covered soil, snow-covered soil, and soil covered by both vegetation and snow. The incoming shortwave and longwave radiation, the ambient air temperature, humidity, and wind speed, and the precipitation rate at the current time step are supplied by the atmospheric model. The energy and moisture budgets of each subarea are calculated separately within CLASS, and the surface fluxes are averaged over the grid cell and passed back to the atmospheric model.

The soil profile is divided into three horizontal layers, of thicknesses 0.10, 0.25 and 3.75 m. The texture of each layer and the overall depth to bedrock are derived from the global dataset assembled by Webb, Bartlein, Harrison, and Anderson (1993). The hydraulic properties of the soil layers are obtained from the soil texture using relationships developed by Cosby, Hornberger, Glapp, and Ginn (1984). The layer temperatures and liquid and frozen moisture contents are carried as prognostic variables and are stepped forward in time using the fluxes calculated at the top and bottom of each layer. Energy fluxes are obtained from the solution of the surface energy balance, expressed as a function of the surface temperature and solved by iteration. The soil albedo and thermal properties vary with texture and moisture content. Moisture fluxes are determined using classic Darcy theory in the case of drainage and capillary rise and after the method of Mein and Larson (1973) in the case of infiltration. If the surface infiltration capacity is exceeded, water is allowed to pond on the surface up to a maximum depth which varies by land cover. Continental ice sheets are modelled in the same way as bare soil, using the thermal properties of ice instead of soil minerals. Snow is modelled as a fourth, variable-depth soil layer with its own separate layer temperature, carried prognostically. Density and albedo vary exponentially with time, from fresh-snow values to specified background values, according to relationships derived from field data. Melting occurs if either the surface temperature or the snow pack layer temperature is projected to rise above 0°C. In this case, the excess energy is used to melt part of the snow pack and the temperature is set back to 0°C. Meltwater percolates into the pack and refreezes until the entire layer reaches 0°C, at which point any further melt is allowed to reach the soil surface. Snowmelt decreases

the thickness of the pack until a limiting depth of 0.10 m is reached; after this, the snow pack is assumed to become discontinuous, and a fractional snow cover is calculated by setting the depth back to 0.10 m and employing conservation of mass.

Vegetation properties such as height, leaf area index, albedo and rooting depth are assigned to each vegetation type on the basis of measurements gleaned from the literature. Derived properties such as the shortwave radiation extinction coefficient, the canopy gap fraction, the roughness lengths for heat and momentum, and the annual cycle of leaf area index are determined separately for coniferous trees, deciduous trees, crops, and grass, and are then averaged over the grid cell to define the bulk canopy characteristics. The canopy temperature and the liquid and frozen intercepted water are carried as prognostic variables. The interception capacity is calculated as a function of leaf area index. Stomatal resistance to transpiration is parameterized as a function of incoming shortwave radiation, air vapour pressure deficit, canopy temperature and soil moisture, using functional relationships similar to those presented by Stewart (1988).

3 Experimental design

Simulations using CanAM4 were performed following the specifications outlined by Taylor, Stouffer, and Meehl (2011) for the Atmospheric Model Intercomparison Project (AMIP), which is a subset of CMIP5. A five-member ensemble of CanAM4 simulations was generated by starting each simulation from slightly perturbed initial conditions. Each simulation started on 1 January 1949, spinning up CanAM4 for 1 year, with the analysis period starting on 1 January 1950 and ending 31 December 2009. Monthly mean AMIP sea surface temperature (SST) and sea-ice boundary condition data were used for the lower boundary. These are the same boundary condition data required for the CMIP5 project (Hurrell, Hack, Shea, Caron, and Rosinski, 2008). See Taylor, Williamson, and Zwires (2008) for a description of this dataset. Time varying monthly averaged concentrations for CO₂, CH₄, O₃, N₂O, effective CFC11 and CFC12 were those of the historical period used in the CMIP5 project and extended to 2010 using the Representative Concentration Pathway (RCP4.5) scenario (Moss et al., 2010). Aerosols were interactive in the simulations, with decadal emissions from anthropogenic and biomass burning sources interpolated to monthly time increments for SO₂ and black and organic carbon (Lamarque et al., 2010). Stratospheric volcanic effects are modelled via the use of specified time- and latitude-varying (four latitude bands) stratospheric aerosol optical depth. The data used were those of Sato, Hansen, McCormick, and Pollack (1993), as extended by the Hadley Centre using an exponential decay out to a minimum value of 0.00001 in 1999 and remaining constant thereafter. Therefore, there were effectively no stratospheric volcanic effects after 1999.

Annual mean total solar irradiance variations are specified according to the CMIP5 protocol. These were determined by direct multiple regression of the sunspot and facular time series with a time series of total solar irradiance, as described in Fröhlich and Lean (2004). See <http://solaris.gfdl-potsdam.de/cmip5.php> for details.

Ozone is specified for radiative transfer calculations using the transient, three-dimensional ozone fields recommended for CMIP5 and described in Cionni et al. (2011). For the period of time covered by the historical segment of the ozone database, 1850–2009, a zonally averaged version of the CMIP5 ozone was used in CanAM4 without further modification.

In CanAM4, the vegetation types present over each grid cell are obtained from the Global Land Cover 2000 (GLC2000) global dataset (Bartholomé and Belward, 2005).

4 Comparison between CanAM4 and AGCM3

Given the substantial number of new or improved parameterizations in CanAM4 that were described in previous sections, CanAM4 provides a more physically complete representation of the atmosphere than does AGCM3. For instance, radiation schemes based on correlated- k distribution and McICA are increasingly replacing other, less accurate radiation schemes in climate models (Barker et al., 2008; Oreopoulos et al., 2012). As another example, the parameterization for transient shallow convection by von Salzen and McFarlane (2002) has led to improved representations of low clouds and convective mixing in CanAM4 (von Salzen et al., 2005) and the global climate model ECHAM5-HAM (Isotta, Spichtinger, Lohmann, and von Salzen, 2011). Furthermore, detection and attribution of climate change and climate projections benefit substantially from the introduction of a prognostic aerosol scheme (e.g., Gillett et al., 2012).

Improvements in parameterizations have led to more skilful seasonal predictions. A coupled forecast system based on CanAM4 produces a markedly higher skill than an earlier system that is based on AGCM3 (Merryfield et al., Unpublished manuscript). Compared to AGCM3, simulations with the CanAM4-based forecast system produce a more vigorous El Niño–Southern Oscillation (ENSO) when employing an identical ocean model, which is generally in better agreement with observations.

For a comparison of broad climatological features, CanAM4 and AGCM3 were configured to run AMIP-type simulations similar to those described in Section 3. Given the formulation of AGCM3, mixing ratios of greenhouse gases and the forcings could not be made identical in these simulations. For example, background aerosols in AGCM3 are specified as constants, assuming present-day conditions. Also, variations in solar irradiance and volcanic aerosol are not accounted for in AGCM3. Overall, simulated and observed climatological mean results agree slightly better with observations in CanAM4 than AGCM3 for a range of different atmospheric quantities (Fig. 2). The most apparent difference between CanAM4 and AGCM3 is an increase in

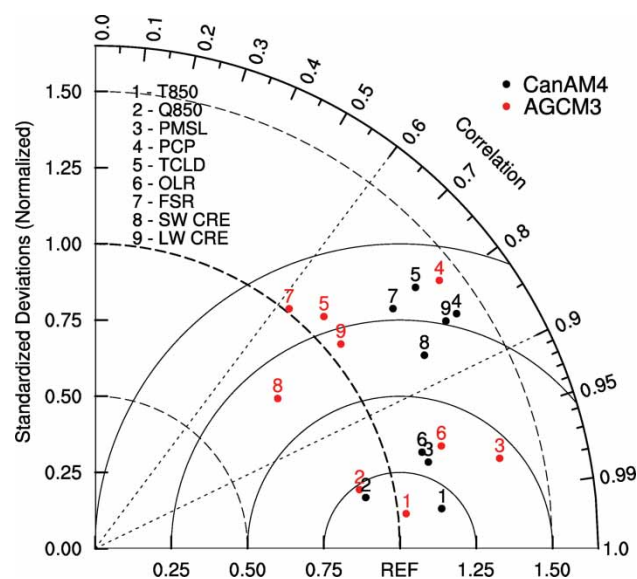


Fig. 2 Taylor diagram (Taylor, 2001) for CanAM4 (black) and AGCM3 (red). The radial coordinate gives the magnitude of the total standard deviation, normalized by the observed value, and the angular coordinate gives the correlation with observations. Numbers indicate model-based datasets compared with observations for global temperature (1) and specific humidity (2) at 850 hPa, mean sea level pressure (3), precipitation (4), total cloud amount (5), outgoing longwave (6) and shortwave (7) radiation at the top of the atmosphere, and shortwave (8) and longwave (9) cloud radiative effects. Mean model results and observations during the time period 2003–08 are used. Observations are from the European Centre for Medium-range Weather Forecasts Re-Analysis (ERA) Interim reanalysis (Dee et al., 2011) for 1–3, the Global Precipitation Climatology Project (GPCP; Adler et al., 2003) for 4, ISCCP D2 (Rossow, Walker, Beuscher, and Roiter, 1996) for 5, and CERES EBAF (Loeb et al., 2009) for 6–9.

spatial variability for many of the simulated quantities, in particular for quantities related to clouds (e.g., for datasets 5, 7, 8, and 9 in Fig. 2). On a global scale, the general increase in spatial variability in the results of CanAM4 tends to counteract improvements in spatial correlation between model results and observations, which leads to similar errors in patterns for both models.

In the following, the analysis of model results will only be based on results from CanAM4 in order to provide a benchmark for studies of clouds and precipitation in the second part of this paper and for other model applications. A more detailed analysis of AGCM3 simulation results is not pursued further given complications associated with differences in forcings and diagnostic capabilities relative to CanAM4. The general performance of this model has been documented in a number of earlier publications.

5 Mean distributions of clouds and precipitation

Clouds cover a wide range of spatial and temporal scales in the atmosphere. The diverse nature of clouds and associated radiative properties poses a major challenge for those seeking to

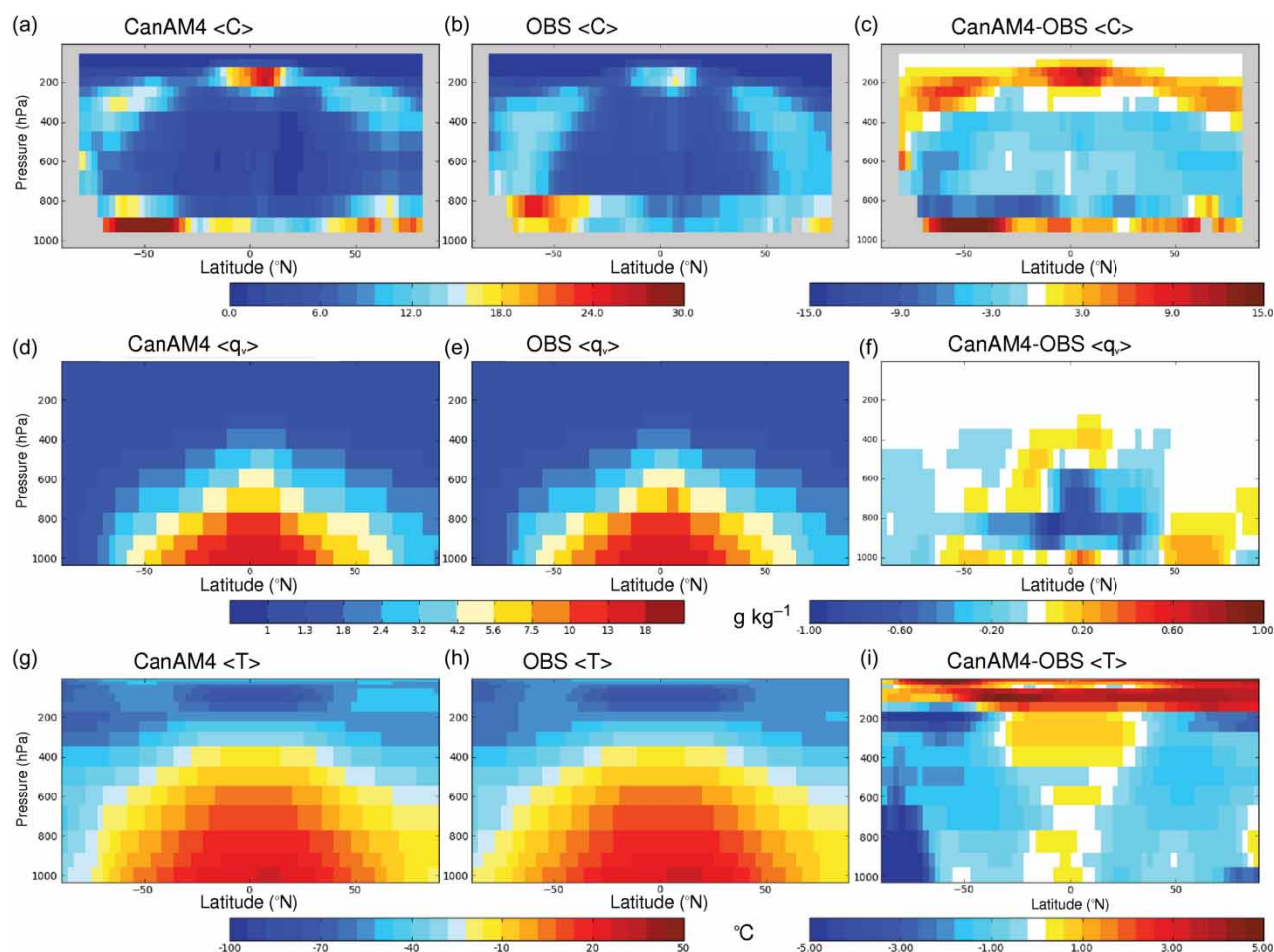


Fig. 3 Simulated and observed zonally and temporally averaged cloud amounts (panels a to c), specific humidity (panels d to f; units: g kg^{-1}), and temperature (panels g to i; units: $^{\circ}\text{C}$) for CanAM4 (left column), CALIPSO-GOCCP satellite observations, and the ERA-interim reanalysis. Differences between model results and observations are displayed in the last column. A logarithmic scale is used for panels d and e. The corresponding time periods are June 2006 to June 2009 for cloud fractions and January 1989 to December 2009 for specific humidities and temperatures. Zonal mean cloud amounts from CALIPSO-GOCCP and COSP were interpolated to pressure levels for the comparison using the simulated geopotential height. Grid points with missing data appear as grey areas. Large temperature anomalies over Antarctica are caused by differences in extrapolation below topography in the different datasets.

understand the role of clouds in climate. Satellite simulators have recently become available for detailed and accurate comparisons between cloud-related quantities from models and satellite-based datasets. For instance, systematic biases in the representation of cloud amounts and optical thickness of different types of clouds in global climate models were identified through application of the International Satellite Cloud Climatology Project (ISCCP; Rossow and Schiffer, 1999) simulator (Klein and Jakob, 1999; Webb, Senior, Bony, and Morcrette, 2001).

Clouds simulated by CanAM4 can be compared with cloud properties from ISCCP, and other satellite platforms, through the use of the CFMIP Observational Simulator Package (COSP; Bodas-Salcedo et al., 2011). To ensure consistency between the diagnosed cloud properties and radiative fluxes, the simulators in COSP were modified so that each satellite simulator used the same subgrid-scale clouds as the CanAM4 radiation (Section 2f).

A comparison of retrieved cloud amount from the GCM-Oriented Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observations (CALIPSO) Cloud Product (CALIPSO-GOCCP; Chepfer et al., 2010) against that simulated by CanAM4 and the COSP is shown in Fig. 3. Zonal mean results for cloud amounts are compared to results for specific humidity and temperature. Broad features in the observations are reasonably well captured by the model. However, amounts of low and mid-level clouds tend to be underestimated in CanAM4 at levels above 925 hPa (Fig. 3c). Similar to CanAM4, most models tend to underestimate amounts of low and mid-level clouds (Zhang et al., 2005), without a known single cause. For CanAM4, underestimates for mid-level clouds in the tropics are related to different causes. Firstly, effects of cumulus congestus clouds are insufficiently parameterized in CanAM4, as is evident from an underestimate of humidity in the tropical free troposphere at mid- and low levels (Fig. 3f). In addition, a more detailed analysis of the

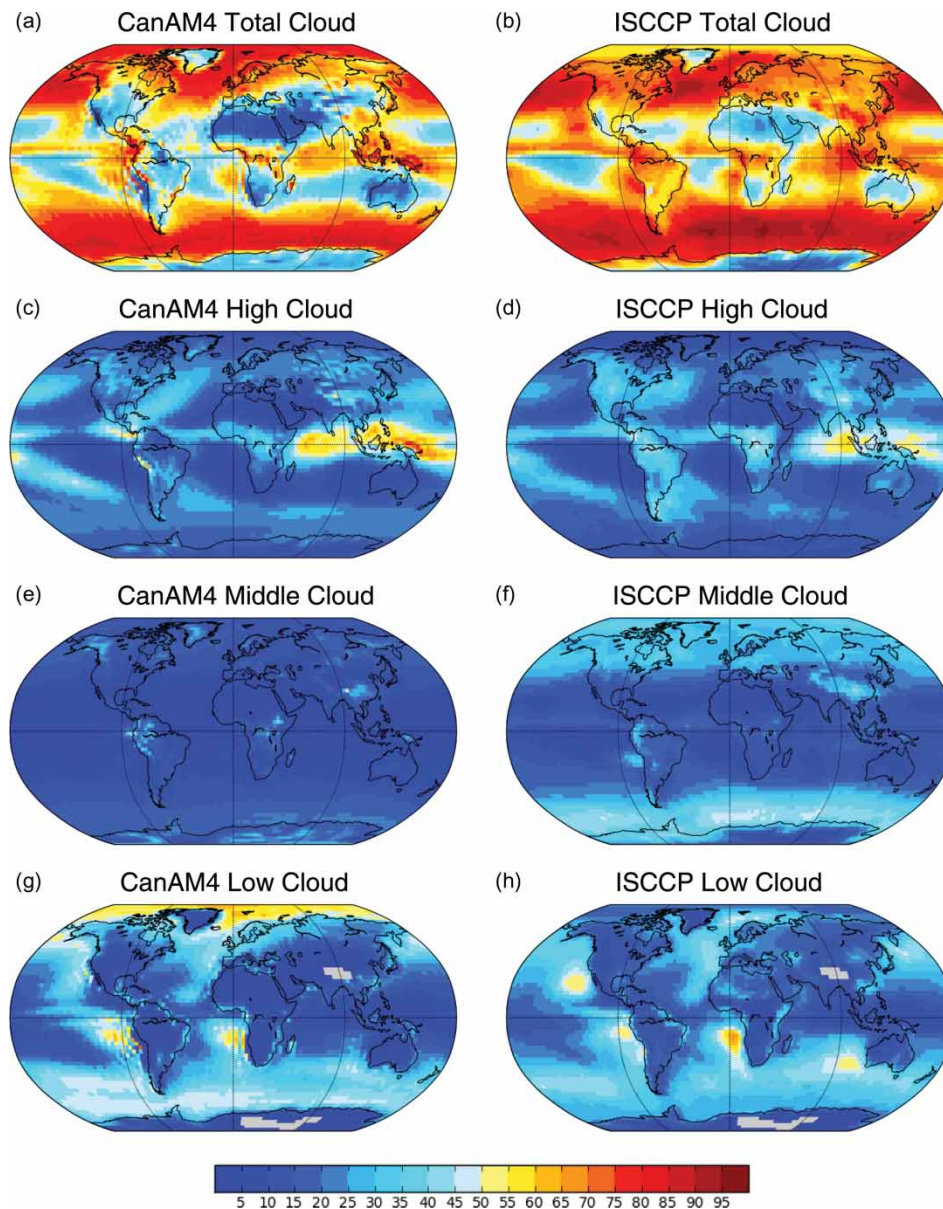


Fig. 4 Mean total cloud fraction for CanAM4 (a) and ISCCP D2 during the time period January 1996 to December 2005. Results are broken down into contributions from high (panels c and d), middle (panels e and f), and low (panels g and h) top clouds using the ISCCP/COSP cloud simulator tool in CanAM4. The corresponding pressure intervals are from 50 to 440 hPa (high), 440 to 680 hPa (middle) and 680 hPa to surface (low). Only the contributions of stratiform clouds and shallow cumulus to total cloud amounts are considered for CanAM4.

cloud data gives evidence that cloud top heights and humidity are underestimated in CanAM4 for marine regions that are mainly affected by stratocumulus clouds. Finally, free-tropospheric extratropical clouds in CanAM4 occur at levels that are too high as is evident from Fig. 3c. The formation of anomalous high-level free-tropospheric extratropical clouds is related to unrealistically cold conditions in the upper free troposphere (Fig. 3i), which points to the general circulation as a potential cause of this bias.

Cloud fractions simulated by the ISCCP/COSP simulator in CanAM4 agree reasonably well with results from

ISCCP D2 for broad climatological features (Fig. 4). Over the subtropical ocean, total cloud fractions are slightly smaller in CanAM4, which can be largely attributed to an overall lack of low-level clouds and humidity. Small-scale spatial inhomogeneities for simulated amounts of clouds (e.g., near the west coast of South America) are caused by numerical truncation which is associated with the spectral transform in regions with sharp gradients in advected quantities (the so-called Gibbs effect). A lack of mid-level clouds largely explains smaller total cloud fractions in the extratropics. Note that differences in polar regions may not

necessarily imply model biases because passive satellite retrievals tend to be less robust over snow and ice found at high latitudes.

Results for cloud fractions in Fig. 4 cannot be directly compared to zonal mean cloud amounts in Fig. 3. Results in Fig. 4 represent vertically projected cloud fractions for cloud tops within different pressure ranges whereas results in Fig. 3 refer to horizontally overlapping cloud amounts at different latitudes and heights. Furthermore, owing to relatively wide pressure ranges that are used for the plots in Fig. 4, vertical shifts in amounts of low and high clouds that are apparent in Fig. 3 do not notably affect results in Fig. 4. In fact, good agreement is found between mean cloud amounts from ISCCP and CALIPSO-GOCCP when both datasets are averaged over the same pressure ranges (not shown).

Amounts and location of clouds in the vertical have considerable implications for radiative transfer in the atmosphere. Combined effects of scattering and absorption of radiation by different types of clouds lead to a large net sink of energy in the atmosphere in the global and annual mean (Trenberth, Fasullo, and Kiehl, 2009). The Clouds and Earth's Radiant Energy System (CERES) Energy Balanced and Filled (EBAF) dataset provides detailed and accurate satellite-based estimates of cloud contributions to atmospheric energy budgets (Loeb et al., 2009). Monthly mean results from CERES EBAF (edition 2) for cloud radiative effects (CRE) were used here. The CRE is calculated as the difference between net all-sky and net clear-sky radiative fluxes at the top of the atmosphere and is sometimes referred to as cloud radiative forcing. Typically the CRE is calculated in models by rerunning the radiative transfer calculations at each time step with the clouds removed and by applying the same large-scale humidity profiles in the calculations. However, in CanAM4 rather than use the large-scale humidity profiles in the radiative transfer calculations we use the clear-sky humidity profiles which are provided by the statistical cloud scheme. This results in CanAM4 computing clear-sky radiative fluxes that are more consistent with satellite-based clear-sky fluxes which are determined from cloud-free footprints (e.g., Sohn, Nakajima, Satoh, and Jang, 2010) and by extension more consistent CREs.

The global and annual mean averaged net CRE for March 2000 to December 2007 is -20.6 W m^{-2} for CERES and -23.1 W m^{-2} for CanAM4. This can be compared with other estimates for the present-day net CRE from ISCCP FD (-23.6 W m^{-2} ; Zhang, Rossow, Lacis, Oinas, and Mishchenko, 2004) and other global climate models (-23.3 W m^{-2} ; Meehl et al., 2007). As mentioned in Section 2, the magnitude of the simulated global mean net CRE depends on the values of several parameters in parameterizations in CanAM4, which are subject to considerable, often unknown, uncertainty. Consequences of parameter choices for simulated climate, clouds, and precipitation have been addressed in numerous studies (e.g., Cole et al., 2011; Sanderson, 2011; Scinocca and McFarlane, 2004; von Salzen et al., 2005).

There is good overall agreement between observed and simulated patterns of the net CRE from CERES EBAF and CanAM4, respectively (Figs 5a and 5b). Similar to CERES EBAF, CanAM4 produces a negative net CRE in the relatively cloudy extratropics and coastal stratus regions. On the other hand, slightly positive forcings are found for regions that are covered by bare ground, ice, or snow.

A breakdown of the net CRE into shortwave (SW) and longwave (LW) components gives evidence for characteristic contributions of low and high clouds to the net CRE (Fig. 5). Observed and simulated SW CREs are dominated by contributions from highly reflective low and mid-level clouds in extratropical and coastal stratus regions (see Fig. 4). Absorption and re-emission of infrared radiation by high-level clouds largely explains the results for the LW CRE. Substantial compensation of negative SW CREs by positive LW CREs is found in regions with deep cloud layers in the convectively active tropics. There is good agreement for global mean results (SW CRE -48.4 W m^{-2} for CanAM4, -47.2 W m^{-2} for CERES EBAF; LW CRE 25.3 W m^{-2} for CanAM4, 26.6 W m^{-2} for CERES EBAF).

Regional differences between results from CanAM4 and observations are similar to biases in several other models (e.g., Williams and Webb, 2009, suppl. material). For instance, the SW CRE tends to be underestimated for stratus clouds in coastal areas in eastern portions of ocean basins, consistent with biases for cloud fractions (Fig. 4). Maxima in LW CRE are unrealistically shifted from the Amazon region to central America and from the eastern to the western tropical Pacific Ocean in CanAM4. The consistency of these differences with biases in the fractions of low and high clouds points to model shortcomings related to parameterizations for clouds and convection or their interactions with other processes in the model.

It is worthwhile to note that the overall good agreement between simulated and observed SW CRE is partly a consequence of compensating biases in the model. Cole et al. (2011) found that cloud-mean albedo for clouds located at low and mid-levels in CanAM4 are larger than those observed by CERES. This is attributable to CanAM4 simulating cloud optical depths, via large liquid water paths, that are too large for these cloud types. These large optical depths are then partly compensated for by cloud fractions that are too small. Similar biases have been found in other models (Zhang et al., 2005) although Klein et al. (2012) have shown that biases are less pronounced in CMIP5 models compared with earlier versions of the same models, including CanAM4.

There is good agreement for mean precipitation from CanAM4 and observations from GPCP during the time period January 1979 to December 2007. Given that precipitation tends to be associated with deep cloud layers with high cloud tops, results for precipitation and LW CRE consistently point to common biases in the representation of clouds in CanAM4, including the shift of clouds from the Amazon

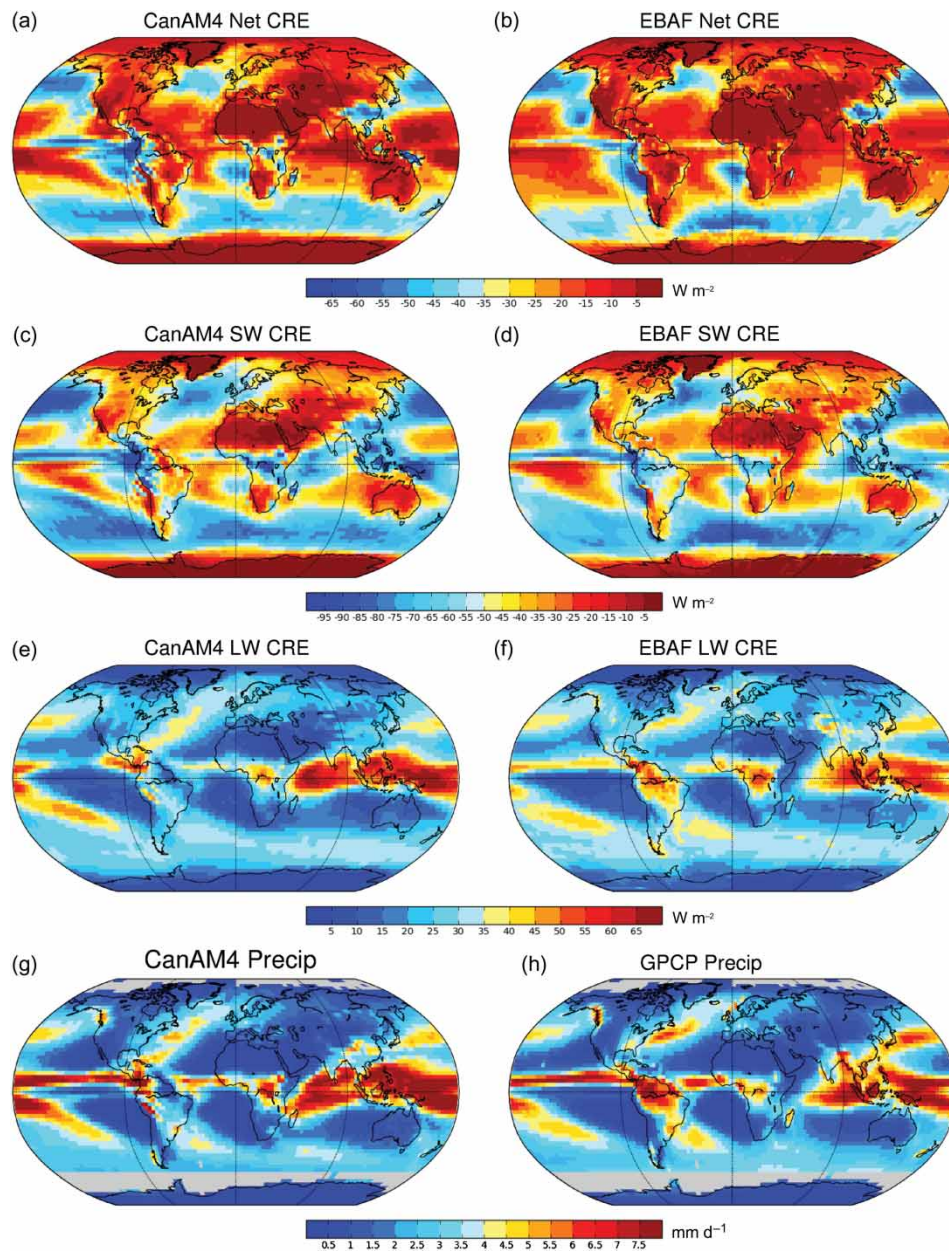


Fig. 5 Mean net, SW, and LW cloud radiative effects from simulations with CanAM4 (left column, panels a, c and e) and observations from CERES EBAF (right column, panels b, d and f) for March 2000 to December 2007. Mean precipitation for CanAM4 (g) and GPCP (h) for January 1979 to December 2007. Grid points with missing data appear as grey areas. For the comparisons, monthly mean observations were first regridded to match the grid used in CanAM4 and accounting for missing data points, as appropriate. Subsequently, results were averaged over the time period for which datasets overlap in time.

region to central America (see Fig. 4). Some of the biases in mean distributions for CRE and precipitation are also associated with biases in the responses of clouds to variations in temperatures, as addressed in the second part of this paper.

6 Summary and conclusions

Parameterizations for clouds, radiation, and other physical processes in the fourth generation Canadian Atmospheric Global Climate Model (CanAM4) have been described.

Numerous changes to parameterizations have resulted in a substantial improvement in model functionality between CanAM4 and a previous version of the model (AGCM3). For instance, previously missing effects of shallow convection, cloud microphysical processes, and aerosol life cycles were added in CanAM4. A noticeable difference between these versions of the model is a larger variability of cloud-related results in CanAM4 which is associated with an increased number and complexity of prognostic parameterizations for clouds and aerosols in CanAM4.

Comparisons of the results of CanAM4 with satellite-based observations for clouds and precipitation give evidence for overall realistic climatological mean results. There is a high consistency between biases in cloud radiative effects and biases in cloud amounts. Biases in the vertical distribution of the clouds apparently play a more subtle role in biases in cloud radiative effects. Biases for clouds and humidity in the tropical free troposphere are likely related to insufficient mixing from cumulus congestus clouds in CanAM4. Low biases in high cloud amounts over the Amazon region in CanAM4 are associated with underestimates in precipitation and biases in cloud radiative effects. Results in other regions also yield consistent biases in mean cloud amount, precipitation, and cloud radiative effects (e.g., the eastern North Atlantic and Indian Ocean).

Improvements to parameterizations for convection and the introduction of a scheme for turbulent kinetic energy are planned in the future to address shortcomings in the simulations of clouds.

Results for clouds are further analyzed in the second part of this paper, which addresses responses of clouds and precipitation to short-term variations in temperature and atmospheric stability.

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Appendix A: Mass fixer

CanAM4 includes a method that locally corrects for any residual changes in globally integrated tracer amounts which result from transport of the hybrid variable in the model. As described in Section 2b, advection of a hybridized instead of a physical variable necessarily leads to sources or sinks of global tracer mass. Different methods for tracer mass correction have been tested. According to the method currently used in CanAM4, the corrected tracer mass mixing ratio,

after the application of the mass fixer, is given by

$$q_{\text{corr}}(t) = q(t)[1 + (c - 1)f], \quad (\text{A1})$$

where $q(t)$ is the initial tracer mass mixing ratio before the correction. The variable c is the ratio of the predicted (i.e., correct) to the initially simulated globally integrated tracer mass,

$$c = \frac{\int \int \int \left[q(t - \Delta t) + \Delta t \left. \frac{dq}{dt} \right|_{\text{phys}} \right] dV}{\int \int \int q(t) dV},$$

where t is the current time and Δt is the model time step. $dq/dt|_{\text{phys}}$ is the time rate of change for the tracer mass mixing ratio between the current and previous time step owing to non-transport processes (i.e., from calculations in the physics part of the model). Non-conservation of global mass from transport calculations implies that $c \neq 1$ in general.

In Eq. (A1), f is used to modulate the magnitude of the mass correction. In AGCM3, $f = 1$ was chosen so the same scaling factor is applied in all grid cells (McFarlane et al., 2006). However, the approach in CanAM4 is to vary the magnitude of the mass correction locally in such a way that the correction tends to be stronger in grid cells that experience large net physical sources or sinks of tracer mass compared to grid cells with weaker sources or sinks. Therefore, corrections for tracer mass are typically much smaller in the stratosphere than in the troposphere for tracer fields that are dominated by emission and removal in the troposphere. In practice, this approach ensures that the impact of the mass correction on local tracer budgets is small relative to the impact of physical processes. In detail,

$$f = (1 - g)k \left| \frac{dq}{dt} \right|_{\text{phys}} + g,$$

with

$$k = \frac{\int \int \int q(t) dV}{\int \int \int q(t) \left| \frac{dq}{dt} \right|_{\text{phys}} dV}$$

$$g = \begin{cases} 0, & \text{if } c \geq 1, \\ \max \left[\min \left(\frac{1 - \varepsilon - k \left| \frac{dq}{dt} \right|_{\text{phys,max}}}{1 - k \left| \frac{dq}{dt} \right|_{\text{phys,max}}}, 1 \right), 0 \right], & \text{if } c < 1, \end{cases}$$

where $|dq/dt|_{\text{phys,max}}$ is the maximum of the absolute tendency that occurs for all grid cells. With this approach, $q_{\text{corr}} \geq q\varepsilon$, with $\varepsilon = 10^{-9}$. The lower bound for q_{corr} ensures that the tracer mass mixing ratio is large enough in the variable transformation according to Eq. (1). Furthermore, $f = 1$ for $c = \varepsilon$.

Appendix B: Overlap of precipitation with clouds

Precipitation that is produced within a layer of the atmosphere may partly overlap with clouds in grid cells below that layer. The overlap between the precipitation flux and the clouds is variable in the vertical and is parameterized according to the maximum-random overlap rule (Geleyn and Hollingsworth, 1979). The fraction of a grid cell which is affected by precipitation is given by

$$C_{k \rightarrow l}^p = 1 - (1 - C_k^p) \prod_{m=k+1}^l \frac{1 - \max(C_{m-1}^p, C_m^p)}{1 - C_{m-1}^p}. \quad (\text{B1})$$

Here, k refers to the topmost grid cell and l the lowest grid cell at the bottom of the layer $k \rightarrow l$, with $k < l$; C_m^p is the local fraction of the cloud which is affected by precipitation,

$$C_m^p = \begin{cases} C_m, & \text{if } P_{m-1} > 0 \\ 0, & \text{otherwise} \end{cases},$$

where P_{m-1} is the local precipitation flux between grid cells $m-1$ and m . Therefore, it is usually true that $C_m^p < C_m$; if

not the entire atmospheric column is affected by precipitation. This approach ensures that only clouds that produce rain will actually contribute to the rain flux.

It is further assumed that $C_{k \rightarrow l}^p$ can be decomposed into a cloudy-sky and a clear-sky component,

$$C_{k \rightarrow l}^p = C_{k \rightarrow l}^{p,\text{cld}} + C_{k \rightarrow l}^{p,\text{clr}}.$$

$C_{k \rightarrow l}^{p,\text{clr}}$ is determined under the assumption of maximum overlap between the precipitation flux and local cloud fraction in each grid cell, so that

$$C_{k \rightarrow l}^{p,\text{clr}} = C_{k \rightarrow l}^p - C_l^p.$$

For simplicity, precipitation fluxes are assumed to be horizontally homogeneous on all scales within the cloudy- and clear-sky components of grid cells.

Separate equations according to the approach in Eq. (B1) are solved for rain and snow.

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